SQL SUBQUERIES

DBMS Study Notes | Concepts, Types, Syntax & Solved Examples

Learning objective: Understand how to write a query inside another SQL query and use its result in the outer query.

Beginner-friendly notes for postgraduate DBMS preparation

1. What is a Subquery?

A **subquery** (also called an inner query or nested query) is a query written inside another SQL statement. The query containing it is called the **outer query**. The result of the inner query is used by the outer query.

Example: Find employees whose salary is greater than the average salary.

```
SELECT employee_name, salary
FROM Employee
WHERE salary > (
    SELECT AVG(salary)
    FROM Employee
);
```

The inner query calculates the average salary. The outer query displays employees whose salary is greater than that average.

2. Sample table used in examples

emp_id	employee_name	department	salary	manager_id
1	Anu	IT	50000	NULL
2	Binu	HR	35000	1
3	Ciya	IT	60000	1
4	Deepa	Sales	40000	2
5	Ebin	HR	30000	2

3. Types of Subqueries

Subqueries can be classified based on the number of values or rows they return, and whether they depend on the outer query.

A. Single-row subquery

Returns only one row (usually one value). Operators such as =, >, <, >= and <= can be used.

```
SELECT employee_name, salary
FROM Employee
WHERE salary > (
    SELECT AVG(salary)
    FROM Employee
);
```

B. Multiple-row subquery

Returns more than one row. Common operators include IN, ANY and ALL.

```
SELECT employee_name
FROM Employee
WHERE department IN (
    SELECT department
    FROM Employee
    WHERE salary > 50000
);
```

The inner query returns department values for employees earning more than 50000. IN checks whether each outer value is present in that result.

C. Multiple-column subquery

Returns more than one column. The outer query compares a group of columns with the values returned by the inner query. The number and order of compared columns must correspond.

```
SELECT employee_name, department, salary
FROM Employee
WHERE (department, salary) IN (
    SELECT department, MAX(salary)
    FROM Employee
    GROUP BY department
);
```

This finds the highest-paid employee or employees in each department in databases that support row-value comparisons.

D. Correlated subquery

A correlated subquery refers to a column from the outer query. It is evaluated using each candidate row from the outer query (the database optimizer may transform its execution).

```
SELECT e.employee_name, e.salary, e.department
FROM Employee e
WHERE e.salary > (
    SELECT AVG(e2.salary)
    FROM Employee e2
    WHERE e2.department = e.department
);
```

This displays employees earning more than the average salary of their own department.

E. Non-correlated subquery

A non-correlated subquery does not refer to the outer query. It can be evaluated independently of the outer query.

```
SELECT employee_name
FROM Employee
WHERE salary > (
    SELECT AVG(salary)
    FROM Employee
);
```

4. Operators used with Subqueries

IN

Checks whether a value matches any value in the list returned by the subquery.

```
SELECT employee_name
FROM Employee
WHERE department IN (
    SELECT department FROM Employee WHERE salary > 50000
);
```

NOT IN

Checks whether a value is not present in the subquery result. Be careful: if the subquery returns NULL, NOT IN can produce unexpected results. Use NOT EXISTS when appropriate.

```
SELECT employee_name
FROM Employee
WHERE department NOT IN (
    SELECT department FROM Employee WHERE salary > 50000
);
```

EXISTS

Returns TRUE if the subquery returns at least one row. It is often used with correlated subqueries.

```
SELECT d.department
FROM Department d
WHERE EXISTS (
    SELECT 1
    FROM Employee e
    WHERE e.department = d.department
```

```
);
```

ANY / SOME

The comparison is TRUE if it is true for at least one value returned by the subquery. SOME is a synonym for ANY in standard SQL.

```
SELECT employee_name, salary
FROM Employee
WHERE salary > ANY (
    SELECT salary FROM Employee WHERE department = 'HR'
);
```

ALL

The comparison must be TRUE for every value returned by the subquery.

```
SELECT employee_name, salary
FROM Employee
WHERE salary > ALL (
    SELECT salary FROM Employee WHERE department = 'HR'
);
```

5. Subquery in different SQL clauses

A subquery can be used in WHERE, HAVING, FROM, and in some cases SELECT. Support and exact syntax can vary slightly by database system.

Subquery in WHERE

```
SELECT employee_name
FROM Employee
WHERE salary > (SELECT AVG(salary) FROM Employee);
```

Subquery in HAVING

```
SELECT department, AVG(salary) AS avg_salary
FROM Employee
GROUP BY department
HAVING AVG(salary) > (
    SELECT AVG(salary) FROM Employee
);
```

Subquery in FROM (derived table)

```
SELECT dept_summary.department, dept_summary.avg_salary
FROM (
    SELECT department, AVG(salary) AS avg_salary
    FROM Employee
    GROUP BY department
) AS dept_summary
WHERE dept_summary.avg_salary > 40000;
```

Subquery in SELECT

```
SELECT employee_name,
       (SELECT AVG(salary) FROM Employee) AS company_avg
FROM Employee;
```

6. Subquery vs JOIN

Subquery	JOIN
A query nested inside another query.	Combines rows from two or more tables.
Often useful when filtering based on an aggregate or a set of values.	Often useful when displaying related columns from multiple tables.
Can be correlated or non-correlated.	Uses join conditions such as ON.

Subquery	JOIN
May sometimes be rewritten as a JOIN, depending on the query.	May sometimes be rewritten as a subquery, depending on the query.

7. Important points to remember

• A subquery is enclosed in parentheses.
• The inner query is generally written inside the outer query.
• A single-row subquery can use comparison operators such as =, > and <.
• A multiple-row subquery commonly uses IN, ANY, ALL or EXISTS.
• A correlated subquery refers to a column from the outer query.
• A subquery used in FROM is commonly called a derived table and generally requires an alias.
• Avoid NOT IN if the subquery may return NULL; consider NOT EXISTS.

8. Exam questions with answers

1. Define subquery.
A subquery is a query written inside another SQL query. Its result is used by the outer query.

2. What are the inner query and outer query?
The inner query is the nested query. The outer query is the query that contains and uses the inner query.

3. What is a correlated subquery?
A correlated subquery refers to a column from the outer query and is evaluated in relation to the outer row being considered.

4. Differentiate single-row and multiple-row subqueries.
A single-row subquery returns one row, while a multiple-row subquery returns more than one row. Single-row comparisons commonly use =, > or <; multiple-row queries commonly use IN, ANY or ALL.

5. What is the purpose of EXISTS?
EXISTS tests whether the subquery returns at least one row.

6. Differentiate ANY and ALL.
ANY requires the comparison to be true for at least one returned value. ALL requires it to be true for every returned value.

7. Can a subquery be used in the FROM clause?
Yes. It acts as a derived table and generally needs a table alias.

8. What is the difference between a JOIN and a subquery?
A JOIN combines rows from tables based on a condition. A subquery nests one query inside another and uses the inner result in the outer query.

9. Practice questions

Write SQL queries for the following:

1. Display employees earning more than the company's average salary.
2. Display employees working in departments where at least one employee earns more than 50000.
3. Display employees earning more than the average salary of their department.
4. Display departments that have at least one employee.
5. Display employees earning more than every employee in the HR department.

Answer key / hints:
1. WHERE salary > (SELECT AVG(salary) ...).
2. Use IN with a subquery selecting department.
3. Use a correlated subquery comparing department values.
4. Use EXISTS with a correlated subquery.
5. Use > ALL with a subquery selecting HR salaries.

10. Quick revision table

Term	Meaning
Subquery	Query inside another query
Single-row	Returns one row
Multiple-row	Returns multiple rows
Multiple-column	Returns more than one column
Correlated	References outer query columns
Non-correlated	Runs independently of outer query
IN	Matches any value in a returned set
EXISTS	Tests whether at least one row exists
ANY	Comparison true for at least one value
ALL	Comparison true for every value

Study tip: Practise tracing the inner query first. Write down its result, then use that result to evaluate the outer query. This makes nested SQL questions much easier.